

Ghost Gutzwiller for the Half-Filled Hubbard Dimer by Direct Minimization (No Impurity Problem): Equations and Implementation of `not_imp_gga_dimer.py`

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Purpose of this note. `gga_dimer.py` solves the ghost Gutzwiller (gGA) equations of the Hubbard dimer by the iterative embedding cycle of `code.tex`: an impurity Hamiltonian $H_{\text{imp}}(V, \lambda^c)$ is diagonalized at every iteration. `not_imp_gga_dimer.py` is an *alternative* solver for *the same problem*: no impurity Hamiltonian, no V^I , no $\lambda^{I,c}$ and no fixed-point loop appear. Following Tagliente, Pasqua and Fabrizio (arXiv:2607.27156, `NotImpurity.pdf`, Eqs. 7 and 9–12), the variational energy functional is minimized *directly* over the impurity wavefunctions $|\phi^I\rangle$ and the quasiparticle Slater determinant, subject to the density constraints. The notation is that of `code.tex` ($N_{\text{eff}} = 1 + N_g$, R^I , Δ^{II} , D^I , $\Delta^{I,\text{bb}}$, ρ , λ^I , $S(\Delta)$, $\mathbf{I}$, X_{ab} , ...); equations of `code.tex` are cited by name, not by number.

1 The problem

Model. Exactly as in `code.tex`: the half-filled two-site Hubbard dimer with hopping $-t$ and local Hamiltonian $H_{\text{loc}} = U n_{\uparrow} n_{\downarrow} - \mu (n_{\uparrow} + n_{\downarrow})$, $\mu = U/2$ (it differs from the $\frac{U}{2}(n-1)^2$ of the

reference paper by the constant $U/2$ per site, which is kept so that energies are directly comparable with `gga_dimer.py`). $N_{\text{phys}} = 1$ physical orbital per fragment, $N_{\text{eff}} = 1 + N_g$ quasiparticle (bath) orbitals per fragment and spin; N_g is even, so N_{eff} is odd.

Symmetry class. As in `gga_dimer.py`: paramagnetic (spin-restricted), the two fragments $I = 0, 1$ *not* assumed equal.

- The impurity wavefunction $|\phi^I\rangle$ is a *real* vector in the half-filled, spin-balanced sector $N_{\uparrow} = N_{\downarrow} = (1 + N_{\text{eff}})/2$ of the embedding Fock space (dimension $4^{1+N_{\text{eff}}}$), a sector of dimension

$$m = \binom{1 + N_{\text{eff}}}{(1 + N_{\text{eff}})/2}^2 \quad (m = 36 \text{ for } N_g = 2). \quad (1)$$

All operators entering below conserve $N_{\uparrow}$ and $N_{\downarrow}$ separately, so their restriction to this sector is exact (no leakage).

- The quasiparticle Slater determinant $|\psi_*\rangle$ has one 1-RDM $\rho \in \mathbb{R}^{2N_{\text{eff}} \times 2N_{\text{eff}}}$ (per spin, the same for both), a rank- N_{eff} projector: half filling, i.e. N_{eff} of the $2N_{\text{eff}}$ levels are occupied (the filling rule of `code.tex`).

Blocks of ρ are written ρ^{IJ} (ρ^{II} is Δ_{qp}^{II} of `code.tex`).

2 The variational functional

Energy (Eq. 11 of the reference). Specialized to the dimer and summed over spin and over the two bonds,

$$E(\rho, \phi^0, \phi^1) = -4t (R^0)^T \rho^{01} R^1 + \sum_{I=0,1} \langle \phi^I | H_{\text{loc}} | \phi^I \rangle. \quad (2)$$

The factor 4 is 2 (spin) $\times 2$ (the two ordered fragment pairs), the same factor as in the variational energy of `code.tex`. Here R^I is *not* an independent variable: it is the function $R(\phi^I)$ of Eqs. 9–10 of the reference,

$$R^I = S(\Delta^I)^{-1} D^I, \quad \Delta^I \equiv \mathbf{I} - \Delta^{I,\text{bb}}, \quad S(\Delta) = \sqrt{\Delta(\mathbf{I} - \Delta) + \varepsilon \mathbf{I}}, \quad (3)$$

with the two expectation values over the impurity state (spin averages)

$$\Delta_{ab}^{I,\text{bb}} = \frac{1}{2} (\langle d_{a\uparrow}^\dagger d_{b\uparrow} \rangle + \langle d_{a\downarrow}^\dagger d_{b\downarrow} \rangle)_{\phi^I}, \quad D_a^I = \frac{1}{2} (\langle d_{a\uparrow}^\dagger c_\uparrow \rangle + \langle d_{a\downarrow}^\dagger c_\downarrow \rangle)_{\phi^I}. \quad (4)$$

These are the same $\Delta^{I,\text{bb}}$ and D^I as in the impurity section of `code.tex`; $S(\Delta)$ is the shifted square root of `code.tex` ($\varepsilon = 10^{-9}$ by default, added to the eigenvalues of $\Delta(\mathbf{I} - \Delta)$). Since $\Delta \mapsto \mathbf{I} - \Delta$ leaves $\Delta(\mathbf{I} - \Delta)$ unchanged, $S(\mathbf{I} - \Delta^{I,\text{bb}}) = S(\Delta^{I,\text{bb}})$, i.e. Eq. (3) is the reference's $R = Q S^{-1}$ with $S^2 = \Delta_\phi(\mathbf{I} - \Delta_\phi)$, $\Delta_\phi = \Delta^{I,\text{bb}}$, $Q = D$.

Constraints (Eq. 7 of the reference).

$$\langle \phi^I | \phi^I \rangle = 1, \quad \rho^{II} = \mathbf{I} - \Delta^{I,\text{bb}}(\phi^I) \quad (I = 0, 1), \quad (5)$$

the second being the requirement that the local density matrix of the Slater determinant equal the particle-hole transform of the bath density matrix of the impurity state.

The functional F (Eq. 12) and the Lagrangian actually used. Eq. 12 of the reference is $F = E - E_* \langle \psi_* | \psi_* \rangle - E_\phi \langle \phi | \phi \rangle + \text{Tr}\{\hat{\lambda}[\mathbf{I} - \hat{\Delta}_\phi - \hat{\Delta}_*]\}$: energy, two normalizations, and a single matrix multiplier $\hat{\lambda}$ for the density constraint. With our spin factors and eliminating the two normalizations (see below), the saddle point of F is the stationary point of

$$\mathcal{L} = E(\rho, \phi^0, \phi^1) + \sum_I 2 \text{Tr} \left[\lambda^I (\mathbf{I} - \Delta^{I, \text{bb}}(\phi^I) - \rho^I) \right], \quad (6)$$

where λ^I is the same λ^I as in the quasiparticle Hamiltonian $H_{\text{qp}} = \begin{bmatrix} -\lambda^0 & -tR^0R^{1T} \\ -tR^1R^{0T} & -\lambda^1 \end{bmatrix}$ of `code.tex`. Compared with the Lagrangian of `code.tex` (which also carries V^I , $\lambda^{I,c}$ and the independent variables R^I, Δ^I), the constraints associated with V^I ($D^I = S(\Delta^I)R^I$) and with $\lambda^{I,c}$ ($\Delta^I + \Delta^{I, \text{bb}} = \mathbf{I}$) have been *substituted*: $R^I = R(\phi^I)$ and $\Delta^I = \mathbf{I} - \Delta^{I, \text{bb}}$. This is the simplification of the direct approach.

3 Variables and parametrization

Unknowns. The unconstrained (redundant) variables are

$$z = (x^0, x^1, X), \quad x^I \in \mathbb{R}^m, \quad X \in \mathbb{R}^{2N_{\text{eff}} \times N_{\text{eff}}}, \quad (7)$$

with the two parametrizations

$$\phi^I = \frac{x^I}{\|x^I\|}, \quad \rho = X(X^T X)^{-1} X^T. \quad (8)$$

The first automatically enforces $\langle \phi^I | \phi^I \rangle = 1$ (so the norm multipliers E_ϕ never appear); the second is the orthogonal projector on the column space of X , a rank- N_{eff} projector for any full-rank X (so $|\psi_*\rangle$ is automatically a half-filled Slater determinant and E_* never appears). The redundancy is a gauge: $x^I \rightarrow c x^I$ and $X \rightarrow XA$ ($A \in GL(N_{\text{eff}})$) leave everything invariant. The number of variables is $n_z = 2m + 2N_{\text{eff}}^2$ ($= 90$ for $N_g = 2$).

Operator matrices (precomputed once). From the sparse Jordan–Wigner operators of `gga_dimer.py` (build `impurity_ops`, mode order $c_\uparrow, c_\downarrow, d_{0\uparrow}, d_{0\downarrow}, \dots$) the following $m \times m$ real matrices are built by restriction to the sector:

$$\begin{aligned} H_{\text{loc}}^s &= \text{sector}(H_{\text{loc}}), \\ \mathcal{Y}_a &= \text{sector}(d_{a\uparrow}^\dagger c_\uparrow + c_\uparrow^\dagger d_{a\uparrow} + d_{a\downarrow}^\dagger c_\downarrow + c_\downarrow^\dagger d_{a\downarrow}), \\ B_{ab} &= \frac{1}{2} \text{sector}(d_{a\uparrow}^\dagger d_{b\uparrow} + d_{a\downarrow}^\dagger d_{b\downarrow}) \quad \text{symmetrized in } (a, b), \end{aligned} \quad (9)$$

together with $n_\uparrow n_\downarrow$ and $n_\uparrow + n_\downarrow$ for the observables. For a real ϕ (so $\langle d^\dagger c \rangle = \langle c^\dagger d \rangle$) the expectation values of Eqs. (4) become plain quadratic forms:

$$\Delta_{ab}^{I, \text{bb}} = \phi^T B_{ab} \phi, \quad D_a^I = \frac{1}{4} \phi^T \mathcal{Y}_a \phi, \quad E_{\text{loc}}^I = \phi^T H_{\text{loc}}^s \phi. \quad (10)$$

(The $\frac{1}{4}$ is $\frac{1}{2}$ from the spin average times $\frac{1}{2}$ from $\mathcal{Y}_a$ containing both $d^\dagger c$ and its transpose, for each spin.) This is the same evaluation as in `impurity_solve` of `gga_dimer.py`, except that the state ϕ is now a *variable* instead of the ground state of H_{imp} .

4 Objective, constraints and analytic gradients

Everything is differentiated analytically (checked against central finite differences by `--check-gradients`: agreement $\sim 2 \times 10^{-10}$ for both E and the constraints).

Evaluation of E (energy). For each fragment: $\phi^I, \Delta^{I,\text{bb}}, D^I$ from Eqs. (8), (10); diagonalize $\Delta^I = \mathbf{I} - \Delta^{I,\text{bb}} = W \text{diag}(d_i) W^T$; $m_i = d_i(1 - d_i) + \varepsilon$, $f_i = m_i^{-1/2}$; $R^I = W[f \circ (W^T D^I)]$; then Eq. (2). (A valid 1-RDM has all $d_i \in [0, 1]$, so up to rounding $m_i > 0$; the shift ε covers rounding of d_i to slightly negative values.)

Gradient with respect to ρ and X .

$$dE = \text{Tr}[G_\rho d\rho] + \dots, \quad G_\rho = \begin{pmatrix} 0 & -2t R^0 R^{1T} \\ -2t R^1 R^{0T} & 0 \end{pmatrix}, \quad (11)$$

and, from $d\rho = P_\perp dX Y X^T + X Y dX^T P_\perp$ with $Y = (X^T X)^{-1}$, $P_\perp = \mathbf{I} - \rho$,

$$\frac{\partial E}{\partial X} = 2 P_\perp G_\rho X Y. \quad (12)$$

Gradient with respect to R^I and through $R(\phi^I)$. The partial derivatives with respect to the (dependent) vectors R^I are

$$g_{R^0} = -4t \rho^{01} R^1, \quad g_{R^1} = -4t (\rho^{01})^T R^0. \quad (13)$$

Write $R = f(M)D$ with $M = \Delta(\mathbf{I} - \Delta) + \varepsilon \mathbf{I}$ and $f(m) = m^{-1/2}$. In the eigenbasis of Δ , $\tilde{u} = W^T g_R$, $\tilde{v} = W^T D$, the Daletskii–Krein derivative of $f(M)$ is the divided-difference matrix

$$\mathcal{F}_{ij} = \frac{f(m_i) - f(m_j)}{m_i - m_j} = -\frac{1}{s_i s_j (s_i + s_j)}, \quad s_i = \sqrt{m_i}, \quad (14)$$

which is written in the cancellation-free form on the right (valid also for $m_i = m_j$). Since $dM_{ij} = (1 - d_i - d_j) d\Delta_{ij}$ in this basis and $d\Delta = -d\Delta^{I,\text{bb}}$,

$$g_D = f(M) g_R = W (f \circ \tilde{u}), \quad \Gamma = \text{sym}[(\tilde{u} \tilde{v}^T) \circ \mathcal{F}], \quad g_{\text{bb}} = -W [\Gamma \circ (1 - d_i - d_j)] W^T. \quad (15)$$

(This is the same structure as the function `gradS` of `code.tex`, here applied as an adjoint.) Collecting the three routes by which ϕ^I enters (through $\Delta^{I,\text{bb}}$, through D^I , and through E_{loc}),

$$g_\phi = 2 \sum_{ab} (g_{\text{bb}})_{ab} B_{ab} \phi + \frac{1}{2} \sum_a (g_D)_a \mathcal{Y}_a \phi + 2 H_{\text{loc}}^s \phi, \quad \frac{\partial E}{\partial x^I} = \frac{1}{\|x^I\|} (g_\phi - (\phi \cdot g_\phi) \phi), \quad (16)$$

the last step being the projection through $\phi = x/\|x\|$.

Constraints and Jacobian (constraints, constraint_jac). The independent entries ($a \leq b$) of the two symmetric matrix equations of Eq. (5),

$$c_{ab}^I(z) = (\rho^{II} + \Delta^{I,\text{bb}} - \mathbf{I})_{ab} = 0, \quad a \leq b, \quad I = 0, 1 \quad (17)$$

give $n_c = N_{\text{eff}}(N_{\text{eff}} + 1)$ equations ($= 12$ for $N_g = 2$). Their gradients are

$$\frac{\partial c_{ab}^I}{\partial x^I} = \frac{1}{\|x^I\|} (\mathbf{1} - \phi \phi^T) 2B_{ab} \phi, \quad \frac{\partial c_{ab}^I}{\partial X} = 2 P_{\perp} G^{(I,ab)} X Y, \quad (18)$$

where $G^{(I,ab)}$ is the symmetric matrix that has $\frac{1}{2}$ at positions $(a, b), (b, a)$ (or 1 at (a, a) if $a = b$) inside the I -th diagonal block and zero elsewhere. (For $N_g = 2$ the constraint Jacobian has full rank 12 at the solutions examined, backward and forward.)

5 Minimization

Algorithm. The constrained problem

$$\min_z E(z) \quad \text{subject to} \quad c(z) = 0 \quad (19)$$

is solved by sequential quadratic programming (`scipy.optimize.minimize`, `method="SLSQP"`, `ftol` = 10^{-13} , up to 1000 iterations), passing the analytic gradient of E and the analytic Jacobian of c . A run is accepted as *feasible* if $\max |c| < 10^{-8}$; among several starts the lowest-energy feasible result is kept. No impurity Hamiltonian is ever diagonalized in the minimization.

Starting points.

1. *Random start* (`random_start`): x^I with independent Gaussian components; X = the occupied orbitals of H_{qp} built from the default cold-start $R^0 = R^1 = (0.9, 0.05, 0.065, \dots)$, $\lambda = 0$ of `gga_dimer.py`, plus $0.05 \times$ Gaussian noise.
2. *From an (R, λ) guess* (`start_from_R_lambda`; files `R.in`, `L.in` in the format of `gga_dimer.py`): X = the N_{eff} lowest eigenvectors of $H_{\text{qp}}(R, \lambda)$; $\Delta^{II} = \rho^{II}$, V^I from the Eq. 7 of `code.tex` and $\lambda^{I,c}$ from its Eq. 8; then the lowest 16 eigenstates of the sector matrix $H_{\text{loc}}^s + \sum_a V_a^I \mathcal{Y}_a + \sum_{ab} \lambda_{ab}^{I,c} 2B_{ab}$ are examined and the one whose bath density best satisfies $\Delta^{I,\text{bb}} = \mathbf{I} - \Delta^{II}$ is taken as x^I . (This single diagonalization is used *only* to construct the starting ϕ^I ; choosing among the low-lying eigenstates matters for the “forward” seed, whose impurity ground state has swapped ghost occupations and would otherwise violate the constraints by ≈ 0.97 , after which SLSQP restores feasibility far away, at $E \approx -0.80$.)

6 Recovery of the multipliers and consistency checks

At a solution the multipliers of Eq. (6) are recovered, and the point is checked against the equations of the embedding approach of `code.tex` (`analyse`).

Stationarity conditions. $\partial \mathcal{L} / \partial \rho = 0$ on the manifold of projectors and $\partial \mathcal{L} / \partial \phi^I = 0$ on the unit sphere give, with $K = \begin{bmatrix} 0 & -tR^0 R^{1T} \\ -tR^1 R^{0T} & 0 \end{bmatrix}$ the hopping part of H_{qp} and $\Lambda = \text{diag}(\lambda^0, \lambda^1)$,

$$[K - \Lambda, \rho] = 0 \quad (\text{i.e. } H_{\text{qp}} = K - \Lambda, \rho \text{ an eigenprojector}), \quad (20)$$

$$\nabla_{\phi} E - 4 \sum_{ab} \lambda_{ab}^I B_{ab} \phi^I = \mu_I \phi^I \quad (I = 0, 1). \quad (21)$$

Both are *linear* in the $2 N_{\text{eff}}(N_{\text{eff}} + 1)/2$ independent entries of λ^0, λ^1 . They are solved *jointly* by linear least squares. (Using Eq. (21) alone is not sufficient: when a ghost is inert the vectors

$B_{ab}\phi$ are linearly dependent, its minimum-norm solution is not unique and need not make ρ an eigenprojector of H_{qp} ; with the joint system the rank is 12/12 for $N_g = 2$.) The residuals of the joint system are reported as “KKT: ϕ stationarity” and “KKT: $[H_{\text{qp}}, \rho]$ ”.

Quasiparticle energies. With the recovered λ^I , $H_{\text{qp}}(R, \lambda)$ is built and diagonalized; its $2N_{\text{eff}}$ eigenvalues are the quasiparticle energies `eqp` of the scans of `code.tex`.

Check against the embedding equations. With $R^I = R(\phi^I)$ and the recovered λ^I , V^I and $\lambda^{I,c}$ are computed from Eqs. 7 and 8 of `code.tex`, and the residual

$$\|(H_{\text{imp}}^s - \langle \phi^I | H_{\text{imp}}^s | \phi^I \rangle) \phi^I\|_{\infty}, \quad H_{\text{imp}}^s = H_{\text{loc}}^s + \sum_a V_a^I \mathcal{Y}_a + \sum_{ab} \lambda_{ab}^{I,c} 2B_{ab}, \quad (22)$$

is reported (“ $H_{\text{imp}}\phi = E\phi$ ”). A small value shows that ϕ^I is an eigenstate of the embedding Hamiltonian of `code.tex` built from the solution, i.e. that the solution of the direct minimization also solves the embedding equations. The observables $Z^I = R^{IT}R^I$, double occupancy and n_{phys} are evaluated from Eqs. (10).

7 Second-order test

For a stationary point with full-rank constraints, a strict local minimum of the constrained energy requires the Hessian of the Lagrangian to be positive definite on the tangent space of the constraints. `hessian_analysis` computes it as follows.

1. Multipliers μ of the constraints from $\nabla E + J^T \mu = 0$ (least squares), $J = \partial c / \partial z$; residual reported.
2. Hessian H of $E + \mu \cdot c$ with respect to all n_z variables by central differences (step $h = 10^{-6}$) of the analytic gradient $\nabla E + J(z)^T \mu$, symmetrized.
3. Tangent space = null space of J (dimension $n_z - n_c = 78$ for $N_g = 2$); eigenvalues of the reduced Hessian $Z_t^T H Z_t$.

The redundancies of the parametrization give exact zero modes: 2 (norms of x^I) + N_{eff}^2 ($X \rightarrow XA$) + $N_{\text{eff}}(N_{\text{eff}} - 1)$ (rotations of the ghost basis of the two fragments), i.e. 17 for $N_g = 2$. A strict minimum has all other 61 eigenvalues positive.

8 Results ($U/t = 2$, $N_g = 2$) and status

The two $U/t = 2$ solutions discussed in `code.tex` (“backward” and “forward”, seeds in `examples/`) are used as tests.

	backward	forward
E_{var}/t (this code)	-3.1256614043	-3.1249999925
E_{var}/t (gga_dimer.py)	-3.1256614043	-3.1250000
$Z^0 = Z^1$	0.99991	0.9375
$\max c $	6×10^{-15}	8×10^{-15}
KKT: $\phi / [H_{\text{qp}}, \rho] / H_{\text{imp}}\phi = E\phi$	$3 \times 10^{-7} / 8 \times 10^{-9} / 1 \times 10^{-7}$	$6 \times 10^{-7} / 6 \times 10^{-8} / 3 \times 10^{-7}$
Hessian: negative / zero / positive	0 / 17 / 61	0 / 20 / 58
smallest positive eigenvalue	2.90	0.0060

Table 1: Solutions of the direct minimization at $U/t = 2$, $N_g = 2$, started from the (R, λ) seeds of `examples/U2_backward` and `examples/U2_forward` (forward: converged copy). The energies agree with `gga_dimer.py`.

- **Backward** is reproduced to 10^{-12} and is a strict local minimum (the 61 positive eigenvalues are all the non-gauge ones; the softest is 2.90).
- **Forward:** unlike the fixed point of `gga_dimer.py` (where ρ^{II} and $\mathbf{I} - \Delta^{I, \text{bb}}$ differ by ≈ 0.97 with swapped ghost occupations), here ρ^{II} and $\mathbf{I} - \Delta^{I, \text{bb}}$ coincide entry by entry (10^{-12}), both with eigenvalues $\{0, \frac{1}{2}, 1\}$: the constraints of Eq. (5) hold. Multipliers exist (KKT residuals $\lesssim 10^{-6}$). The reduced Hessian has no negative eigenvalue but 3 zero modes in addition to the 17 gauge ones, and its positive eigenvalues are small (0.0060 smallest). Along all 20 zero-mode directions, followed on feasible curves, E never decreases and grows as s^4 (about 3×10^{-6} at $s = 0.2$); thus the second-order test is *inconclusive along those 3 directions*. The origin of these extra flat directions was not investigated. An independent empirical test (random displacements projected back on the constraints, 4000 of them, steps 10^{-3} – 0.1) finds only energy increases growing as s^2 with coefficient $\approx 7 \times 10^{-3}$.
- **Convergence caveat.** With the default tolerance a run started from the forward seed can stop before convergence ($E = -3.12499887$, projected gradient 4×10^{-3} , descent directions with $\Delta E \simeq -8 \times 10^{-5}$ exist); it relaxes to -3.1249999925 with more iterations.

Limitation: random starts. From random starts the minimization always reaches the *higher* boundary minima with inert ghosts, never the ghost-assisted solutions: 120/120 starts give $E = -3.125$ at $U/t = 2$ (versus -3.125661), 30/30 give -10.0 at $U/t = 10$ (versus -10.101314 from `gga_dimer.py`), 30/30 give -6.0 at $U/t = 6$; at $U/t = 4$ the minima are -4.5 and -4.0 . Perturbing the boundary minimum by up to 0.3 in the variables and re-minimizing also returns to it. The ghost-assisted solutions are reached only from a seed close to them (as with the backward seed above). Making the direct minimization find them without a seed (for example by continuation in U , warm-starting (x^I, X) from the previous U) has *not* been done.

9 Usage and correspondence with `gga_dimer.py`

```
python3 not_imp_gga_dimer.py --U <U/t> --Ng <even> [--t 1.0] [--nstarts 10] [--seed 0]
                                [--R-file R.in --lam-file L.in] # seeded start (else random)
                                [--max-iter 1000] [--shift 1e-9] [--quiet]
python3 not_imp_gga_dimer.py --check-gradients [--Ng 2 --U 2] # analytic vs finite diff.
```

It prints Z^I , E_{var}/t , the double occupancy, the constraint violation, the three KKT residuals and `eqp`, and writes `not_imp_Ng*_U*_{R,lambda,lambda_c}.dat` in the layout of `gga_dimer.py`'s `save_results`.

	<code>gga_dimer.py</code> (embedding)	<code>not_imp_gga_dimer.py</code> (direct)
unknowns R^I	R^I, λ^I (and $V^I, \lambda^{I,c}, \Delta^{II}$ derived) output of the impurity solve, $S(\Delta^{\text{new}})^{-1}D$	$x^I(\phi^I)$, $X(\rho)$ function of ϕ^I , Eq. (3)
$V^I, \lambda^{I,c}$	Eqs. 7, 8 of <code>code.tex</code> , needed in every iteration	not used; only in the a-posteriori check
impurity ground state constraint $\rho^{II} = \mathbf{I} - \Delta^{I,\text{bb}}$	diagonalized every iteration <i>not enforced</i> by the convergence test	never (only for a seeded start) enforced ($\max c \sim 10^{-14}$)
solver	damped fixed-point iteration	SLSQP on E with equality constraints
result type	fixed point of the iteration	constrained local minimum (checked)

Table 2: Correspondence between the two solvers.

10 Reconstruction of R and λ from x and X

The unknowns of the direct minimization are only $z = (x^0, x^1, X)$; neither R^I nor λ^I is a variable. Both are reconstructed from a solution in the function `analyse`, in the following order. Let $n_s = N_{\text{eff}}(N_{\text{eff}} + 1)/2$ (the number of independent entries of a symmetric $N_{\text{eff}} \times N_{\text{eff}}$ matrix; $n_s = 6$ for $N_g = 2$).

Step 1: the states.

$$\phi^I = \frac{x^I}{\|x^I\|}, \quad \rho = X(X^T X)^{-1} X^T, \quad \rho^{IJ} = \rho_{[IN_{\text{eff}}:(I+1)N_{\text{eff}}, JN_{\text{eff}}:(J+1)N_{\text{eff}}]}. \quad (23)$$

Step 2: R^I (explicit). R^I is a function of ϕ^I alone, Eqs. (3) and (10):

$$\Delta_{ab}^{I,\text{bb}} = \phi^{IT} B_{ab} \phi^I, \quad D_a^I = \frac{1}{4} \phi^{IT} \mathcal{Y}_a \phi^I, \quad \mathbf{I} - \Delta^{I,\text{bb}} = W \text{diag}(d_i) W^T, \\ R^I = W \left[(d_i(1 - d_i) + \varepsilon)^{-1/2} \circ (W^T D^I) \right]. \quad (24)$$

The matrix ρ plays no role here. $Z^I = R^{IT} R^I$.

Step 3: λ^I (by linear least squares). The multiplier λ^I is not determined by (x, X) through a formula but by the two stationarity conditions of Eq. (6), Eqs. (20) and (21), which are *linear* in the entries of λ^0, λ^1 . Collect them in the unknown vector

$$w = (\lambda_{ab}^0, \lambda_{ab}^1)_{a \leq b} \in \mathbb{R}^{2n_s}. \quad (25)$$

Let $\kappa_{ab} = 1$ for $a = b$ and $\kappa_{ab} = 2$ for $a < b$ (because $\sum_{ab} \lambda_{ab} B_{ab} = \sum_{a \leq b} \kappa_{ab} \lambda_{ab} B_{ab}$), $P_I = \mathbf{1} - \phi^I \phi^{IT}$, $K = \begin{bmatrix} 0 & -tR^0 R^{1T} \\ -tR^1 R^{0T} & 0 \end{bmatrix}$ (with the R^I of Step 2), and let $\Xi^{(I,ab)}$ be the $2N_{\text{eff}} \times 2N_{\text{eff}}$

symmetric matrix with 1 at positions $(IN_{\text{eff}} + a, IN_{\text{eff}} + b)$ and $(IN_{\text{eff}} + b, IN_{\text{eff}} + a)$ (a single 1 if $a = b$) and zero elsewhere. The linear system is

$$\begin{pmatrix} A^{(0)} & 0 \\ 0 & A^{(1)} \\ C^{(0)} & C^{(1)} \end{pmatrix} w = \begin{pmatrix} b^{(0)} \\ b^{(1)} \\ \text{vec}[K, \rho] \end{pmatrix}, \quad (26)$$

a system of $2m + 4N_{\text{eff}}^2$ equations for the $2n_s$ unknowns, with

$$\begin{aligned} A_{\cdot, (ab)}^{(I)} &= 4\kappa_{ab} P_I (B_{ab} \phi^I) \in \mathbb{R}^m, & b^{(I)} &= P_I g_\phi^{(I)}, \\ C_{\cdot, (ab)}^{(I)} &= \text{vec}[\Xi^{(I, ab)}, \rho] \in \mathbb{R}^{4N_{\text{eff}}^2}, \end{aligned} \quad (27)$$

where $g_\phi^{(I)}$ is the gradient of E with respect to ϕ^I at fixed ρ (Eq. (16), including the dependence through $R(\phi^I)$), and the projector P_I removes the norm multiplier μ_I of Eq. (21). The first two block rows are the ϕ^I conditions, the last block row is $[\Xi, \rho]$ -form of Eq. (20) ($[K - \Lambda, \rho] = 0 \Leftrightarrow [\Lambda, \rho] = [K, \rho]$). The system is solved in the least-squares sense (`numpy.linalg.lstsq`, `rcond=10-10`, minimum-norm solution), and λ^I is the symmetric matrix built from the n_s entries of w belonging to fragment I .

- *Why jointly.* The ϕ block alone is rank deficient whenever a ghost is inert (then the vectors $P_I B_{ab} \phi^I$ are linearly dependent), so its minimum-norm solution is not unique and need not satisfy $[K - \Lambda, \rho] = 0$; this happened for the forward solution ($[H_{\text{qp}}, \rho] \approx 0.16$). With the C block included the system has rank $2n_s = 12$ for $N_g = 2$ at the backward and forward solutions examined.
- *Quality check.* If (x, X) is a stationary point, the residual of Eq. (26) vanishes: it is reported as “KKT: ϕ stationarity” (the A rows) and “KKT: $[H_{\text{qp}}, \rho]$ ” (the C rows, recomputed with the final H_{qp}). Values found: $\lesssim 6 \times 10^{-7}$ and $\lesssim 6 \times 10^{-8}$ (Table 1). A large residual means that (x, X) is *not* a stationary point of $\mathcal{L}$, and the λ obtained is then only the best least-squares one.

Step 4: what follows. $H_{\text{qp}} = K - \text{diag}(\lambda^0, \lambda^1)$ is built as in `code.tex`, giving the quasiparticle energies `eqp`; V^I and $\lambda^{I,c}$ follow from Eqs. 7 and 8 of `code.tex` with ρ , R^I , $\Delta^I = \mathbf{I} - \Delta^{I, \text{bb}}$ and λ^I . The arrays R^I , λ^I , $\lambda^{I,c}$ are written by `save_results` in the layout used by `gga.dimer.py` (two rows of N_{eff} numbers for R ; two stacked $N_{\text{eff}} \times N_{\text{eff}}$ blocks for each matrix), so the files have the format of `R.in/L.in`.

Gauge. R^I and λ^I are obtained in the basis defined by x^I (the ghost-orbital basis of the impurity state) and X ; no gauge fixing is applied. Solutions related by a rotation $Q_I \in O(N_{\text{eff}})$ of the ghost basis of fragment I have $R^I \rightarrow Q_I R^I$ and $\lambda^I \rightarrow Q_I \lambda^I Q_I^T$, while E , Z^I and the quasiparticle energies are invariant. (The inverse map, from (R, λ) to a starting (x, X) , is the seeded start described in Section 5.)

References

- [1] A. M. Tagliente, I. Pasqua and M. Fabrizio, *Direct minimization versus iterative embedding in the ghost-Gutzwiller method: a comparative study of magnetism in Mott insulators*, arXiv:2607.27156 (NotImpurity.pdf); Eqs. 7 and 9–12 are the basis of this note.

- [2] `doc/code.tex`: notation and equations of the embedding solver `gga_dimer.py`, used throughout.